

4.1 Data Tables

Excel can perform calculations on data, which you know very well. The fact is, you can also make decisions based on the data: what if this variable were this much more? It turns out that this is actually called “What-If Analysis.” This deals with observing exactly how changes in the input cause changes in the results of a formula.

In Excel, a Data Table is the tool to perform a What-If analysis on a formula. It’s pretty simple: the changes in the results of the formula—caused by changing the values of the input variables—can be tabulated in a Data Table.

At the centre of a What-If analysis lies the formula. Since the Data Table in Excel 2007 can support a maximum of two-variable What-If Analysis, the formula being put to the test needs to meet this condition. There’s no real magic happening here: you can do a What-If Analysis by replacing the input variable and noting the results. But since you have Data Tables, it’s simplified. Just command Excel to make a Data Table, and change what needs to be changed—and the What-Ifs appear all nice and neat in the table.

Now for some rules. The different values that you intend to test your formula with need to be filled in a single column or row, if only one input variable is changing. You can’t have them scattered. If two input variables are changing, the two series of numbers need to be put in a column and a row. Excel has strict rules about the relative positions of the rows and columns and the formula-containing cell, and you need to follow these for the tool to function.

You fire up the Data Table tool from the Data Table Link under the What-If Analysis drop-down in the Data tab.

To make it easier to understand what’s going on, let’s use a simple example involving capital and rate of interest. Say cell B1 contains Capital Amount and cell B2 contains the Rate of Interest. The formula to calculate interest (Capital multiplied by Rate of